

```
!pip install pandas
```

```
in /usr/local/lib/python3.12/dist-packages (2.2.2)
l.26.0 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.0.2)
lateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0.post0
)20.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2026.1)
; in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas) (1
```

```
import pandas as pd
import numpy as np
```

```
print(pd.__version__)
```

```
2.2.2
```

```
#from list
s=pd.Series([1,2,3,4,5])
print(s)
```

```
0    1
1    2
2    3
3    4
4    5
dtype: int64
```

```
s=pd.Series([1,2,3,4,5],index=['a','b','c','d','e'])
print(s)
```

```
a    1
b    2
c    3
d    4
e    5
dtype: int64
```

```
student={
    "name":"Raju",
    "branch":"AIML"
}
se = pd.Series(student)
print(se)
```

```
name      Raju
branch    AIML
dtype: object
```

```
n=pd.Series([10,20,30,40,50])
```

```
s.mean()
```

```
np.float64(3.0)
```

```
s.sum()
```

```
np.int64(15)
```

```
s.max()
```

```
5
```

```
s.describe()
```

	0
count	5.000000
mean	3.000000
std	1.581139
min	1.000000
25%	2.000000
50%	3.000000
75%	4.000000
max	5.000000

```
dtype: float64
```

```
s[s>3]
```

	0
d	4
e	5

```
dtype: int64
```

```
s * 2
```

```
s+10
```

	0
a	11
b	12
c	13
d	14
e	15

```
dtype: int64
```

```
names=pd.Series(['Raju','Rani','Ram'])  
names.str.lower()
```

```
names.str.len()
names.str.startswith('A')
```

0

0 False

1 False

2 False

dtype: bool

```
student={
    "names":["bharu","esther","reena"],
    "branch":["aiml","aiml","aiml"],
}
s=pd.DataFrame(student)
print(s)
```

	names	branch
0	bharu	aiml
1	esther	aiml
2	reena	aiml

```
std=[
    {"names":"bharu","branch":"aiml","age":21},
    {"branch":"reena","branch":"aiml","age":19},
]
s=pd.DataFrame(std)
print(s)
```

	names	branch	age
0	bharu	aiml	21
1	NaN	aiml	19

```
arr=np.random.rand(3,4)
c=pd.DataFrame(arr,columns=['A','B','C','D'])
print(c)
```

	A	B	C	D
0	0.342192	0.734239	0.419091	0.751729
1	0.739403	0.282620	0.416532	0.693973
2	0.590052	0.384840	0.603424	0.649696

```
data={
    'Name':['ravi','hari','siri','ram','Ajay'],
    'Age':[23,25,22,28,30],
    "Score":[88,75,92,81,90],
    "Grade":["A","B","A","A","D"]
}
d=pd.DataFrame(data)
```

```
d.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4 entries, 0 to 3
Data columns (total 4 columns):
```

```
#   Column  Non-Null Count  Dtype
---  -
0   Name    4 non-null      object
1   Age     4 non-null      int64
2   Score   4 non-null      int64
3   Grade   4 non-null      object
dtypes: int64(2), object(2)
memory usage: 260.0+ bytes
```

```
d.describe()
```

	Age	Score
count	4.000000	4.000000
mean	24.500000	84.000000
std	2.645751	7.527727
min	22.000000	75.000000
25%	22.750000	79.500000
50%	24.000000	84.500000
75%	25.750000	89.000000
max	28.000000	92.000000



```
d.columns
```

```
Index(['Name', 'Age', 'Score', 'Grade'], dtype='object')
```

```
d.index
```

```
RangeIndex(start=0, stop=4, step=1)
```

```
d.dtypes
```

```

      0
Name  object
Age   int64
Score int64
Grade object
dtype: object
```

```
d.nunique
```

```
pandas.core.frame.DataFrame.nunique
def nunique(axis: Axis=0, dropna: bool=True) -> Series
```

```
>>> df.nunique(axis=1)
0    1
1    2
2    2
dtype: int64
```

```
d.isnull().sum()
```

```

      0
Name  0
Age   0
Score 0
Grade 0

dtype: int64
```

```
d['Name']
d.Name
d[['Name', 'Score']]
```

	Name	Score
0	ravi	88
1	hari	75
2	siri	92
3	ram	81

```
d.loc[0]
d.loc[0:2]
d.loc[0:2, 'Name': 'Score']
```

	Name	Age	Score
0	ravi	23	88
1	hari	25	75
2	siri	22	92

```
d.iloc[0]
d.iloc[0:2]
d.iloc[0:2, 0:2]
d.iloc[-1]
```

```
      3
Name  ram
Age   28
Score 81
Grade A
```

dtype: object

```
d.loc[0, 'Name']
d.iloc[0,0]
```

'ravi'